Santosh Singh

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GitHub ◊ LinkedIn ◊ LeetCode

EDUCATION

Dehradun Institute of Technology — **Dehradun, India**
Bachelor of Technology Computer Science & Engineering — 2023 – 2027

SGRR Public School — **Dehradun, India**
Senior Secondary School (Class XII) — Jul 2021 – Apr 2023
Percentage: 73.45%

OLOHES — **Silvassa, India**
High School (Class X) — May 2013 – Apr 2021
Percentage: 79.67%

EXPERIENCE

Celebal Technologies Pvt. Ltd. — **Jun 2026 – Aug 2026**
Celebal Excellence Intern (CEI) — Data Science

- **Program:** Selected for Celebal Technologies' CEI program, a structured, industry-focused internship combining guided learning, mentorship, and hands-on project work on real-world Data Science use cases.
- **Learning Track:** Undergoing an industry-aligned curriculum to build technical foundations, with exposure to industry-relevant tools, datasets, and mentorship from experienced professionals.
- **Evaluation:** Program concludes with a performance-based assessment and a Certificate of Completion from Celebal Technologies.

PROJECTS

RAG-Based AI Study Tool (GitHub) (Python, FastAPI, React, LangChain, ChromaDB)

- **RAG Architecture:** Designed and deployed a full-stack Retrieval-Augmented Generation (RAG) app that converts uploaded PDFs, DOCX, and PPTX files into AI-generated quizzes, flashcards, and study notes.
- **Retrieval Pipeline:** Built a semantic search pipeline using ChromaDB vector storage and Sentence-Transformer embeddings, orchestrated with LangChain across multiple LLM providers (Mistral AI, Groq, Gemini).

AI-Powered Code Review Agent (GitHub) (Python, FastAPI, Ollama/LLM)

- **Agent Design:** Built an AI code-review agent that uses a locally hosted LLM (Ollama running CodeLlama) to analyze Python code for bugs, security vulnerabilities, performance issues, and best-practice violations.
- **API & Interface:** Exposed the agent through a FastAPI REST API with endpoints for pasted-code and file-based analysis, returning structured reviews with a 0–10 quality score, plus a lightweight web UI.
- **Testing & Privacy:** Wrote unit tests with pytest for core agent logic and kept all inference on-device, so source code never leaves the local machine.

End-to-End House Price Prediction Model (GitHub) (Python, ZenML, MLflow)

- **Pipeline Design:** Built an end-to-end machine learning pipeline to predict house prices, using ZenML to orchestrate reproducible pipeline stages.
- **Experiment Tracking:** Tracked experiments, metrics, and model versions with MLflow across data ingestion, preprocessing, feature engineering, and evaluation.

TECHNICAL SKILLS

Languages: Java, Python.
Frameworks: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, FastAPI, React, LangChain.
Core Concepts: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, Retrieval Augmented Generation (RAG).
Tools: Jupyter Notebook, Git/GitHub, MLflow, ZenML, Docker, GitHub Actions (CI/CD), Ollama.
Relevant Coursework: Data Analysis, Software Engineering, Operating Systems, Algorithms, Artificial Intelligence.

ACHIEVEMENTS

- **LeetCode:** Solved 280+ Data Structures & Algorithms problems (Java), arrays, strings, hash tables, dynamic programming, and binary search.